

HDOC Day-38

Character Analysis and Case Operations on Strings Using C

Aim

To write a menu-driven C program using separate functions to count spaces, digits, and special characters; count the frequency of a given character; toggle the case of each alphabetic character; and exit.

Algorithm

1. Start the program and read a string.
2. Display the four menu options.
3. Accept the user's choice and process it using a switch statement.
4. Traverse the string to count spaces, digits, and special characters.
5. For frequency, accept a character and count its occurrences in the string.
6. For toggle case, convert uppercase letters to lowercase and lowercase letters to uppercase while leaving other characters unchanged.
7. Exit when option 4 is selected.
8. Display an error for invalid choices and repeat the menu until Exit.

C Program

```
#include <stdio.h>

void countCharacters(char str[]);
void characterFrequency(char str[]);
void toggleCase(char str[]);

int main()
{
    char str[200];
    int choice, i = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0')
    {
        if (str[i] == '\n')
        {
            str[i] = '\0';
            break;
        }
        i++;
    }

    do
    {
        printf("\n----- STRING OPERATIONS MENU ----- \n");
        printf("1. Count Spaces, Digits and Special Characters\n");
        printf("2. Count Frequency of a Character\n");
        printf("3. Toggle Case of Each Character\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: countCharacters(str); break;
            case 2: characterFrequency(str); break;
```

```

        case 3: toggleCase(str); break;
        case 4: printf("Exiting program...\n"); break;
        default:
            printf("Invalid choice! Please enter a number from 1 to 4.\n");
    }
} while (choice != 4);

return 0;
}

void countCharacters(char str[])
{
    int i = 0, spaces = 0, digits = 0, special = 0;

    while (str[i] != '\0')
    {
        if (str[i] == ' ')
            spaces++;
        else if (str[i] >= '0' && str[i] <= '9')
            digits++;
        else if (!(str[i] >= 'A' && str[i] <= 'Z') ||
                 (str[i] >= 'a' && str[i] <= 'z'))
            special++;
        i++;
    }

    printf("Number of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

void characterFrequency(char str[])
{
    char ch;
    int i = 0, count = 0;

    printf("Enter the character to search: ");
    scanf(" %c", &ch);

    while (str[i] != '\0')
    {
        if (str[i] == ch)
            count++;
        i++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

void toggleCase(char str[])
{
    int i = 0;
    char toggled[200];

    while (str[i] != '\0')
    {
        if (str[i] >= 'A' && str[i] <= 'Z')
            toggled[i] = str[i] + 32;
        else if (str[i] >= 'a' && str[i] <= 'z')
            toggled[i] = str[i] - 32;
        else
            toggled[i] = str[i];
        i++;
    }

    toggled[i] = '\0';
    printf("Toggled String: %s\n", toggled);
}

```

Test Case Table

Test	Operation	Input	Expected Result
1	Count Characters	Hello 123!	Spaces=1, Digits=3, Special=1

Test	Operation	Input	Expected Result
2	Count Characters	C@UPES 2026	Spaces=1, Digits=4, Special=1
3	Frequency	banana; a	Frequency = 3
4	Frequency	Programming; m	Frequency = 2
5	Frequency	hello; z	Frequency = 0
6	Toggle Case	Hello World	hELLO wORLD
7	Toggle Case	CSE123@Upes	cse123@uPES
8	Invalid Choice	Choice = 5	Invalid choice message
9	Exit	Choice = 4	Program exits

Sample Output

Enter a string: Hello 123!

Enter your choice: 1
Number of spaces = 1
Number of digits = 3
Number of special characters = 1

Enter your choice: 2
Enter the character to search: l
Frequency of 'l' = 2

Enter your choice: 3
Toggled String: hELLO 123!

Enter your choice: 4
Exiting program...

Result

The menu-driven C program was successfully implemented using separate functions, strings, loops, conditional statements, and a switch-case statement. It counts spaces, digits, and special characters, determines character frequency, toggles case, and continues until Exit is selected.